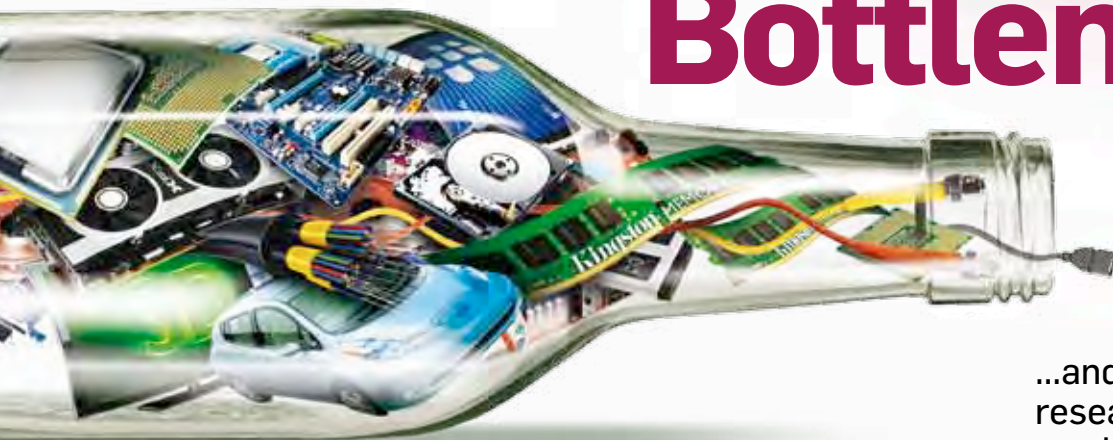




Technology Bottlenecks

Currently Holding Our Future to Ransom...

...and cutting-edge research that is promising a solution

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WH. Auden once wrote in his famous poem, 'As I Walked Out One Evening': "All the clocks in the city began to whirr and chime: O' let not time deceive you, you cannot conquer time." A facet technology seems willing to ignore at this juncture. Several years have passed since the introduction of 1 GHz CPUs, Blu-Ray Discs and even electric cars. And yet, technology seems stuck, unable to progress further into the beyond and achieve the next stage of performance. But it doesn't mean it's not close. Let's take a look at some of the biggest technology bottlenecks currently holding the future to ransom, and the advances coming our way to set us free (without the horrifying consequences that discovering a world conquered by machines would bring).

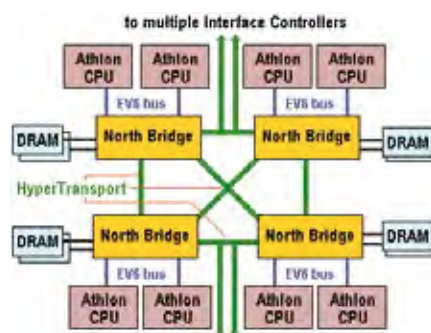
CPU Speed

When processors broke the 1 GHz barrier, it was a significant achievement in computer history. However, over the past few years, CPU speeds have hovered around 2.5 GHz, when you take single-core processors into consideration. Yes, multi-core systems seemed to be the breakthrough we needed, but they carry their own bottlenecks (see

next). Also, despite said breakthrough, a single core still clocks in at around 1.5-2 GHz. So why hasn't processor speeds increased over the years? A prominent factor is transistor size – they're getting smaller, which allows more transistors to be packed onto a single die (thus obeying Moore's Law), but they're not getting faster. CPU transistors these days follow the metal-oxide semiconductor field-effect transistor or MOSFET scaling process, and they

to increased performance. However, MOSFET scaling has its fair share of bottlenecks. The electrical circuit itself faces various issues with higher sub-threshold conduction (which consumes half of the total power consumption), difficulties in scaling down the transistor size beyond a certain point (as Intel has done with its Ivy Bridge chipset, managing a 22 nm size); interconnect capacitance (which is the metal-layer capacitance between various portions of the chip. As it becomes greater, there is an increased delay in the travel of signals through the interconnect, leading to lesser performance), and many more.

Another issue is the "interconnect bottleneck". Integrated circuits are downscaled in size to allow transistors to run at higher frequencies – at the cost of tighter packing of the already dense CPU. This increases parasitic capacitance – a type of capacitance created simply by the proximity of electrical components, which thus causes them to divert from their best possible performance. In more serious scenarios, it can lead to "crosstalk" wherein signals from one circuit meld into another, thus generating inference in operation. There's also the issue of signal propagation



AMD's Hypertransport forms point-to-point links between different components, eliminating the need for a front-side bus.

function as electronic signal switchers. As they become smaller, transistors are supposed to switch faster, which leads